

Chapter → 7

Loops in Python

Sometimes we want to repeat a set of statements in our program. For instance: Print 1 to 1000

Loops make it easy for a programmer to tell the computer which set of instructions to repeat and how!

★ Types of loops in Python

primarily there are two types of loops in python.

- while loops
- for loops

• While loops

Syntax:

`while (condition):` # The block keeps executing until the condition is true

In while loops, the condition is checked first. if it evaluates to true, the body of the loop is executed otherwise not.

if the loop is entered, the process of [condition check & execution] is continued until the condition becomes False.

Quick quiz: write a program to print 1 to 50 using a while loop.

Example:

```
i = 0
while i < 5:                # print "Mahip" - 5 times!
    print("Mahip")
    i = i + 1
```

Note: If the condition never becomes false, the loop keeps getting executed.

Quick quiz: write a program to print the content of a list using while loops.

● For loops

A for loop is used to iterate through a sequence like list, tuple, a string (iterables).

Syntax:

```
l = [1, 7, 8]
```

```
for item in l:
```

```
    print(item)
```

print 1, 7 and 8.

● Range Function in Python

The range() function in Python is used to generate a sequence of numbers.

We can also specify the start, stop and step-size as follows:
range(start, stop, step-size)

step size is usually not used with range()

- An example Demonstrates Range () function
for i in range (0, 7):

```
    print(i)
```

```
# print 0 to 6
```

- For loop with else

An optional else can be used with a for loop if the code is to be executed when the loop exhausts.

Example:

```
l = [1, 7, 8]
```

```
for item in l:
```

```
    print(item)
```

```
else:
```

```
    print("done")
```

● The Break Statement

"break" is used to come out of the loop when encountered. It instructs the program to exit the loop now.

Example:

```
for i in range (0, 10):
```

```
    print(i)
```

```
# this print only 0, 1, 2, 3
```

```
    if i == 3:
```

```
        break
```

● The Continue Statement

"Continue" is used to stop the current iteration of the loop and continue with the next one. It instructs the program to "skip the iterations".

Example:

```
for i in range(4):
```

```
    print("printing")
```

```
    if i == 2:
```

```
        continue
```

```
    print(i)
```

if i is 2, the iteration is skipped.

● Pass Statement

pass is a null statement in Python
It instructs to "do nothing".

Example:

```
l = [1, 7, 8]
```

```
for i in range:
```

```
    pass
```

Practice Set

Chapter → 7

- 1) Write a Program to print multiplication Table of a given number using for loops.
- 2) Write a Program to greet all the person names stored in a list 'l' and which starts with S.
l = ["Mahip", "Soham", "Sudhanshu", "Snehal", "Shikhar"]
- 3) Attempt Problem 1 using while loop.
- 4) Write a program whether given number is prime or not
- 5) Write a program to find the sum of first n natural numbers using while loop.
- 6) Write a program to calculate the factorial of given number using for loop.
- 7) Write a program to print the following star patterns.
*
* * *
* * * * * for n = 5.
- 8) Write a program to print multiplication table of n using in reversed order.